Can past financials predict next-period revenue?

Indian paint companies: 10 ML models and a PyTorch network vs simple rules vs management guidance · Portfolio case study by Jatin Bhalla · October 2026

4.1% vs 9.6%

Annual error. "Same as last year" vs the best ML model (Lasso). The simple rule wins.

3.8% vs 9.8%

Quarterly error. "Same quarter last year x recent growth" vs best ML (Gradient Boosting).

21%

Deep learning error with a clean split (5 seeds). The README's 12% relied on a test-set leak and a lucky run.

3.8% vs 12.8%

Q1 FY2027, a real quarter after the data ends. Guidance-based estimate vs best ML.

Answer in one paragraph. Partly. Revenue in this industry is very "sticky", so past data tells you the *level* and the seasonal pattern very well. That is why even a no-brainer rule is about 4% off. But the ML and DL models built here add **no value on top of that**: every one of them is less accurate than simply copying last year (annual) or the same quarter last year (quarterly). The headline R^2 of 0.96 to 0.99 hides this. What history cannot see is *why* revenue changes: price hikes, raw-material shocks, competition and demand. Management guidance contains exactly that information, which is why it beat every model in the first quarter after the data ends.

1. Problem statement

Research question: "Can we use historical financial and operational data of Indian paint companies to predict their future financial performance, particularly next-period / next-year revenue?" A forecast is useful only if it is **more accurate than the obvious alternatives** that an analyst already has for free: last period's number, and what management says on the earnings call. This report tests the project against both.

2. Data

Four listed companies (Asian Paints, Berger Paints, Kansai Nerolac, Indigo Paints), FY2015 to FY2026, built from **211 official filings** with a source and page for every value (3,082 values, arithmetic-checked). Standalone basis: **45 company-years** and **169 company-quarters**. Inputs are revenue, material cost, EBITDA, PAT, balance sheet and cash flow, plus derived margins, growth rates and ratios. Known breaks: Ind AS (FY2017), GST (FY2018), leases (FY2020), COVID quarters.

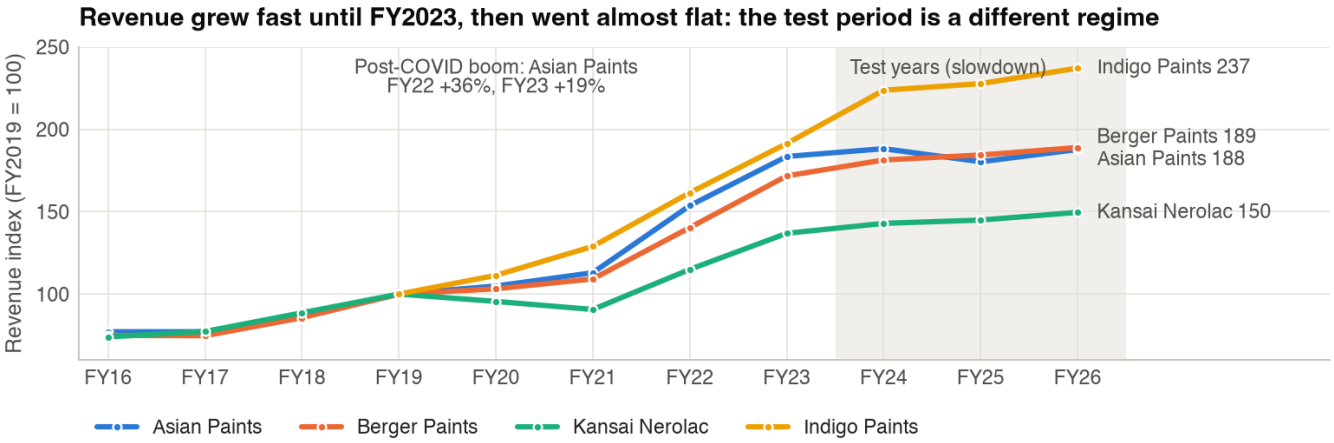


Figure 1. The training years include the post-COVID boom (FY2022 to FY2023). The test years (FY2024 to FY2026) were a slowdown, with Birla Opus entering in February 2024 and a price war. A model that learns "revenue grows about 12% a year" from the training years will over-forecast the test years.

3. Methodology

What the project built:

- **Features:** lags (t-1, t-2, t-4), growth momentum, margins, raw-material ratio, capex/OCF/FCF ratios, ROE/ROA, 4-quarter rolling mean and volatility, quarter dummies and company dummies (19 annual and 21 quarterly features).
- **Target:** next year's revenue (annual) and next quarter's revenue (quarterly), in INR crore.
- **Out-of-time split:** annual train FY2015 to FY2022, test = predictions of FY2024 to FY2026 (12 rows). Quarterly train to FY2023 Q4, test = 44 next-quarter predictions (FY2024 Q2 to FY2026 Q4).
- **10 ML models:** Linear, Ridge, Lasso, ElasticNet, SVR, Random Forest, Gradient Boosting, HistGB, XGBoost, LightGBM. **Deep learning:** PyTorch MLP, 4 hidden layers (128-64-32-16), He init, BatchNorm, Dropout 0.15, Huber loss, AdamW, early stopping.
- **Metrics:** MAE, RMSE, MAPE and R^2.

What this audit added:

- **Naive baselines** on the same test rows: "same as last year", "same quarter last year" and "same quarter last year x latest year-on-year growth". These are the standard yardsticks in forecasting and were missing.
- **Deep learning re-run** with a clean validation year (FY2023) and 5 random seeds.

- **A real out-of-sample test:** Q1 FY2027 (April to June 2026) results were published in July and August 2026, after the dataset ends. All forecasts are compared with them.
- **Management guidance** from the May 2025 and May 2026 earnings calls and presentations, compared with the model forecasts and with what actually happened.

4. Results: ML models vs each other, vs deep learning, vs simple rules

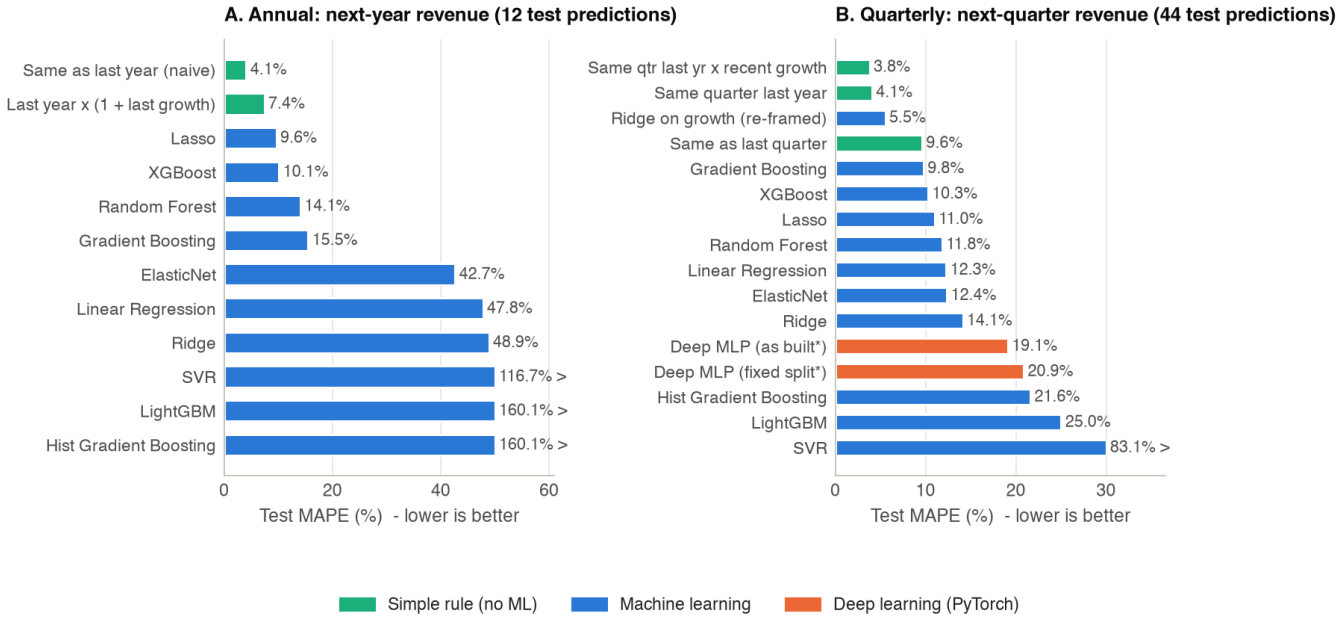


*Figure 2. Test MAPE (average % error) on identical rows. Bars beyond the axis are clipped and marked ">". *Both Deep MLP values are means over 5 random seeds. As built, it used the 44 test quarters for early stopping (range 12.6% to 28.8%); the fixed version validates on FY2023 instead. The DL model was only built for the quarterly task.*

- **Among ML models:** Lasso (annual, 9.6%) and Gradient Boosting (quarterly, 9.8%) were the best. HistGB and LightGBM failed completely on the annual task: with only 29 training rows they cannot make a single split (default minimum of 20 rows per leaf), so they predict one constant number.
- **ML vs deep learning:** ML wins. The Deep MLP averaged 19.1% as built and 20.9% with a clean split, worse than every reasonable ML model. It has 14,177 weights and only 121 training rows, so it cannot learn a stable pattern; results swing a lot between random seeds.
- **Simple rules beat everything:** "same as last year" (4.1%) is 2.3x more accurate than the best annual ML model, and "same quarter last year x recent growth" (3.8%) is 2.6x more accurate than the best quarterly model. Even the "re-framed" Ridge model I built to predict growth instead of crore (5.5%) beat every original model but still lost to the simple rule.

5. Why the numbers looked good, and what went wrong

The R^2 trap. The four companies differ in size by more than 20x (Asian Paints about INR 30,000 crore, Indigo about INR 1,300 crore). Any model that only knows "Asian Paints is big and Indigo is small" scores a high R^2. A "cheat" that predicts each company's average test-period revenue scores $R^2 = 0.999$, higher than any model here. R^2 on pooled companies measures size, not forecasting skill. MAPE per company, compared against a naive baseline, is the honest metric.

- **No baseline.** A 10% error sounded good, but copying last year gives 4%.
- **Regime change.** Trained on years averaging about 12% growth, tested on years with mostly 0% to 5%. Lasso predicted +10% to +14% growth in every test year and over-forecast **11 of 12** rows, by 6 to 18 points.
- **Wrong model picked for the forward forecast.** The notebook ranks by RMSE, so it chose ElasticNet (MAPE 42.7%). ElasticNet predicted **negative** revenue for Indigo in FY2024 (INR -419 crore) and forecasts FY2027 growth of +138% for Indigo and -15% for Nerolac.
- **Missing seasonal anchor.** To predict quarter t+1, the key input is the same quarter last year (lag 3). The model has lags 1, 2 and 4, but not 3.
- **Deep learning leak.** The test set was used as the validation set for early stopping, so the test set influenced training.
- **Too little data, no outside drivers.** 29 annual training rows for 19 features. Inputs are only past financials: nothing on price hikes, crude or TiO2 prices, competition or demand.
- **Smaller issues.** Missing values are filled with medians computed over all years including test years. The trajectory plot draws next period's actual at the current date. README figures do not match the notebook outputs (DL 12.18% vs 17.52%; annual 9.51% vs 9.65%).

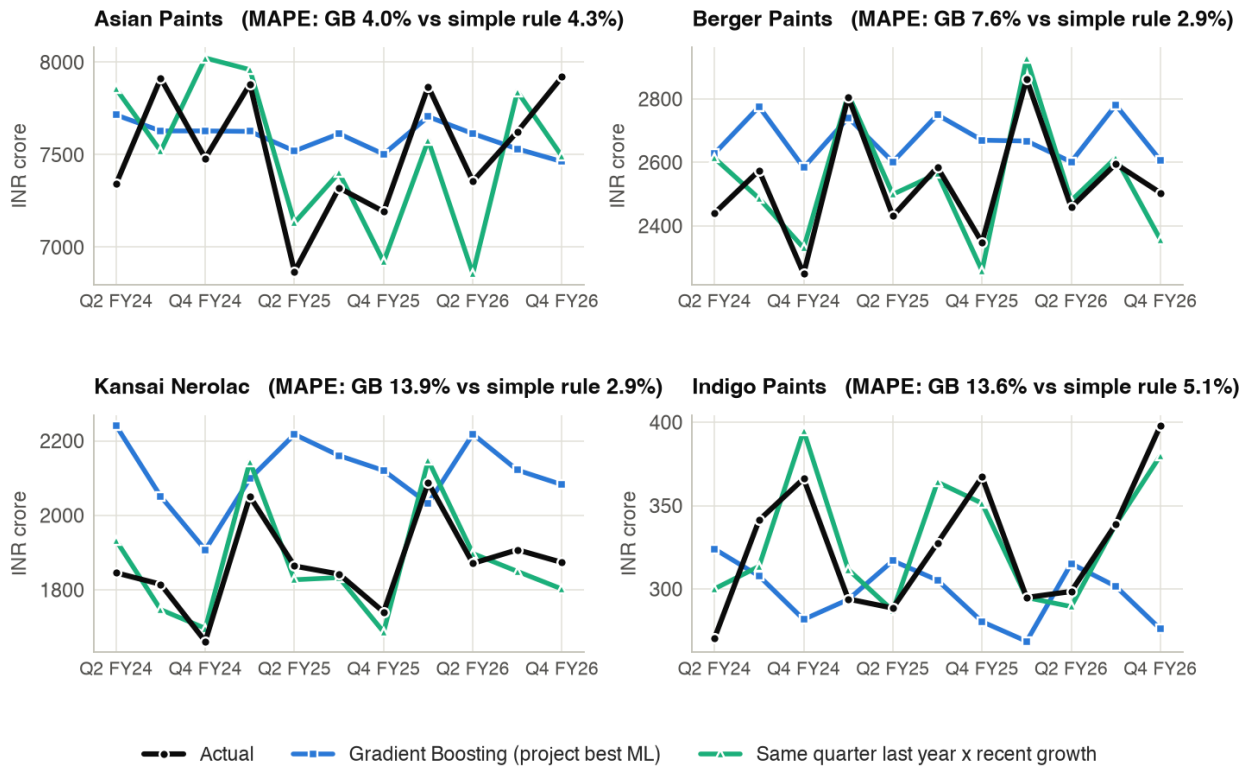


Figure 3. Quarterly test period, plotted at the quarter being predicted. The simple rule (green) follows the seasonal zig-zag; Gradient Boosting (blue) mostly predicts a flat recent average. Its two most important inputs are the 4-quarter rolling mean and current revenue (67% of importance), so it behaves like "same as last quarter" (9.6%).

6. Reality check: model forecasts vs management guidance vs actual results

How guidance works in this sector. Indian paint companies rarely give a revenue number. They guide on **volume growth**, **EBITDA margin bands** and **price actions**, and say whether demand is improving. So the comparison is on direction and size of growth, not on an exact rupee figure.

Company	Actual FY2026	Lasso said	ElasticNet said	"Same as last year"	Management guidance (May 2025)
Asian Paints	+4.1%	+14.4%	+8.4%	0.0%	"Single-digit value growth"; margin 18-20% (actual 20.1%). **Correct.**
Berger Paints	+2.5%	+13.1%	+3.6%	0.0%	Margin 15-17% (actual 16.2%). No revenue number.
Kansai Nerolac	+3.2%	+11.9%	-11.4%	0.0%	No revenue number.
Indigo Paints	+4.1%	+12.4%	+126.2%	0.0%	Demand to normalise by Q2 FY2026 (qualitative; H1 stayed weak).

Table 1. FY2026 back-test (forecast made from FY2025 data). Lasso said about +12% to +14% for everyone because that is the historical average; actual growth was 2% to 4%. Asian Paints management said "single digit" and was right.

Company	What management said in May 2026 (before Q1 FY2027)	Model's FY2027 forecast ElasticNet / Lasso	Q1 FY2027 actual (year on year)
Asian Paints	Decorative **volume growth 8-10%** in FY2027; EBITDA margin 18-20%; price hikes taken in April-May against about 20% input inflation; expects dealers to stock up before hikes.	-1.3% / +13.7%	**+16%** (volume +9%)
Berger Paints	EBITDA margin 15-17%; calibrated price hikes from late March; "gradual recovery" in demand. In August it guided double-digit growth for FY2027.	+4.4% / +13.0%	**+13%**
Kansai Nerolac	EBITDA margin 13-14%; price increases taken (brokers: high single digit overall); no revenue number.	-14.7% / +11.8%	**+10%**
Indigo Paints	Industry price hikes "approximately 12%"; strong April as dealers stocked up; agreed with an analyst on **about 24% growth** (2x an industry growing about 12%).	+138.1% / +12.8%	**+19%**

Table 2. FY2027 outlook. Management flagged price hikes and dealer stocking; the model cannot see either. Lasso's +12% to +14% looks close only because Lasso predicts roughly that every year: it said +10% to +14% for FY2024 to FY2026 and over-forecast 11 of 12 rows.

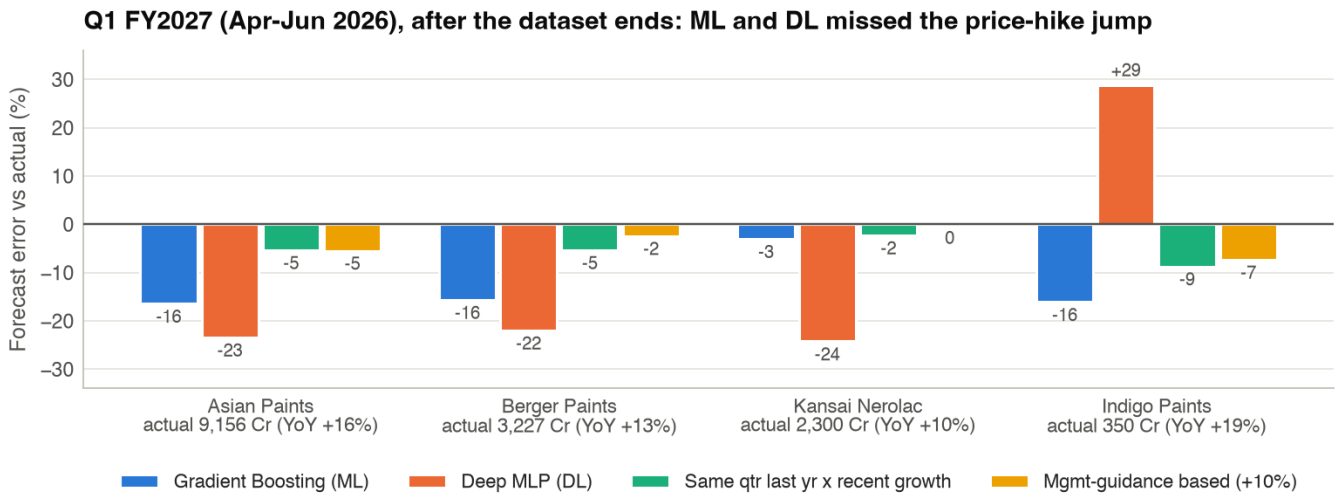


Figure 4. Q1 FY2027 forecast errors (actual standalone revenue from the companies' results; Asian Paints figure is standalone net sales). "Mgmt-guidance based" is my conservative translation of the May 2026 calls into one number: Q1 FY2026 revenue x 1.10 (Asian Paints guided 8-10% volume growth; industry price hikes about 12%). Average error: guidance 3.8%, simple rule 5.4%, Gradient Boosting 12.8%, Deep MLP 24.6%.

7. Conclusion and best model

- **Can the model predict next-period revenue?** It produces numbers that are about 10% off. That is **not** useful, because copying last year (4.1%) or the same quarter last year adjusted for recent growth (3.8%) is far more accurate. In its current form the model has no predictive value beyond the naive rules.
- **Best model today:** for a pure-data forecast, **"same quarter last year x latest year-on-year growth"** (3.8% quarterly). For the annual horizon, "same as last year" (4.1%). Among trained models, the growth-target Ridge (5.5%) was the best and the only one close to the rules. On the first unseen quarter it also did well (about 3.5% average error).
- **ML vs DL:** on this dataset, ML (Gradient Boosting 9.8%) clearly beats DL (about 21%). Deep learning needs far more rows than 4 companies x 12 years can provide.
- **Guidance vs model:** in a stable year guidance and naive rules agree. At a turning point (the April-June 2026 price-hike jump) only guidance-aware forecasts were close. The most useful product is therefore a **hybrid**: a seasonal baseline that management guidance and price actions adjust.

8. How to improve the project

- Report every model against naive baselines (or MASE = model error / naive error) and show MAPE per company, not pooled R^2.
- Predict **growth %** instead of crore; anchor on the same quarter last year; flag or drop the COVID quarters.
- Use **walk-forward validation** (re-train every quarter and predict the next one) instead of one split, and keep a separate validation set for early stopping.
- Add the drivers revenue actually depends on: announced price hikes, crude and TiO2 prices, INR/USD, monsoon, housing sales, a competitor-entry flag, and guided volume growth parsed from earnings-call transcripts (an NLP feature).
- Publish forecast ranges (prediction intervals) rather than single numbers, and choose the final model by MAPE on a validation period, not on the test set.

How to present this in an interview (30 seconds). "I built 10 ML models and a PyTorch network to forecast revenue for India's four listed paint companies from 211 official filings. The headline R^2 was 0.99, but when I benchmarked against naive rules none of them won: the same quarter last year adjusted for recent growth was 2.6x more accurate. I traced it to regime change, an early-stopping leak in the DL model and missing external drivers. I then tested everything on a quarter released after the data ended, where guidance-informed forecasts beat every model. The lesson: always beat the baseline before you trust the model."

Sources. Dataset: company quarterly results and annual reports (see sources.csv). Guidance and Q1 FY2027 results: Indigo Paints Q4 FY2026 earnings-call transcript (25 May 2026, indigopaints.com); Asian Paints Q4 FY2026 and Q1 FY2027 results and call coverage (Yahoo Finance, Investing.com, Upstox, Whalesbook); Berger Paints Q4 FY2026 and Q1 FY2027 coverage (Republic World, Sahi, Free Press Journal); Kansai Nerolac Q4 FY2026 and Q1 FY2027 coverage (Business Standard, PL Capital, Investywise); Asian Paints Q4 FY2025 summary (Quartr, ICICI Securities). Press figures should be checked against exchange filings.